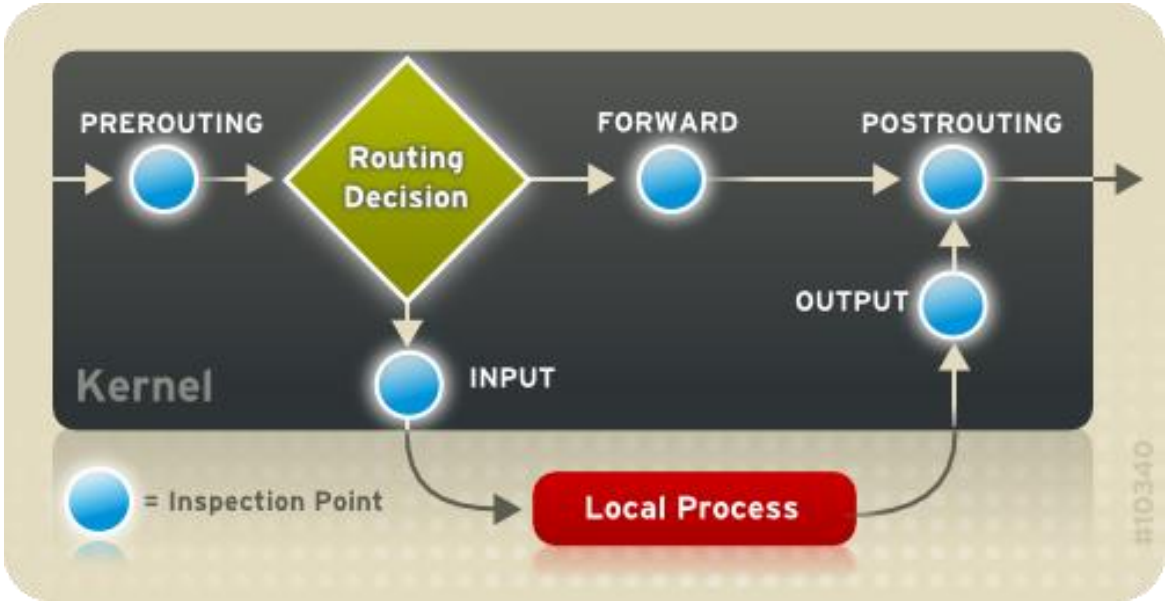


Linux Firewall

Ing. Christian Callegari, PhD

IPTables

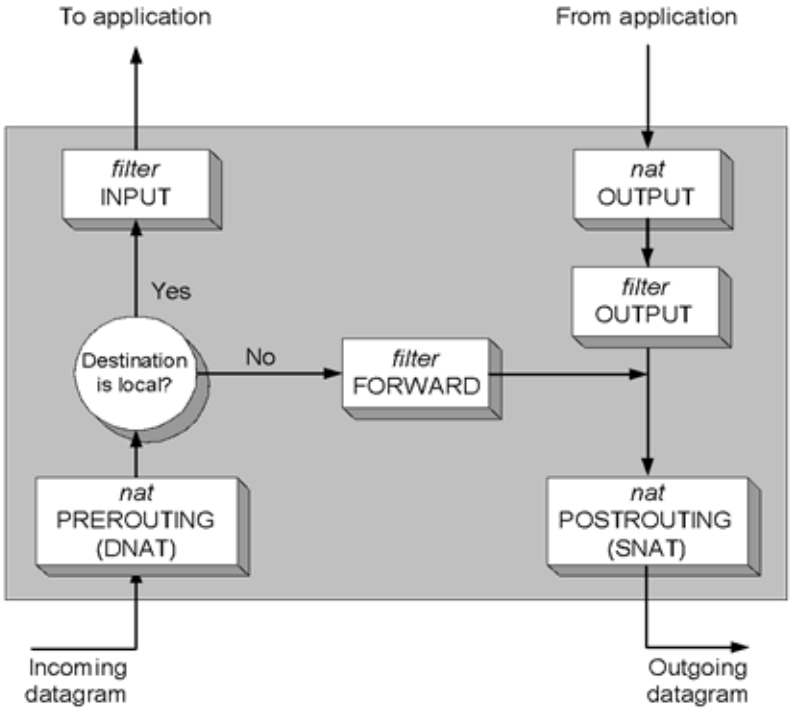
- Linux Firewall kernel module
- From kernel 2.4
- Three main concepts:
  - Tables
    - FILTER
    - NAT
    - MANGLE
  - Chain
    - PREROUTING
    - INPUT
    - FORWARD
    - OUTPUT
    - POSTROUTING
  - Rules



FILTER Table

Made of 3 chains:

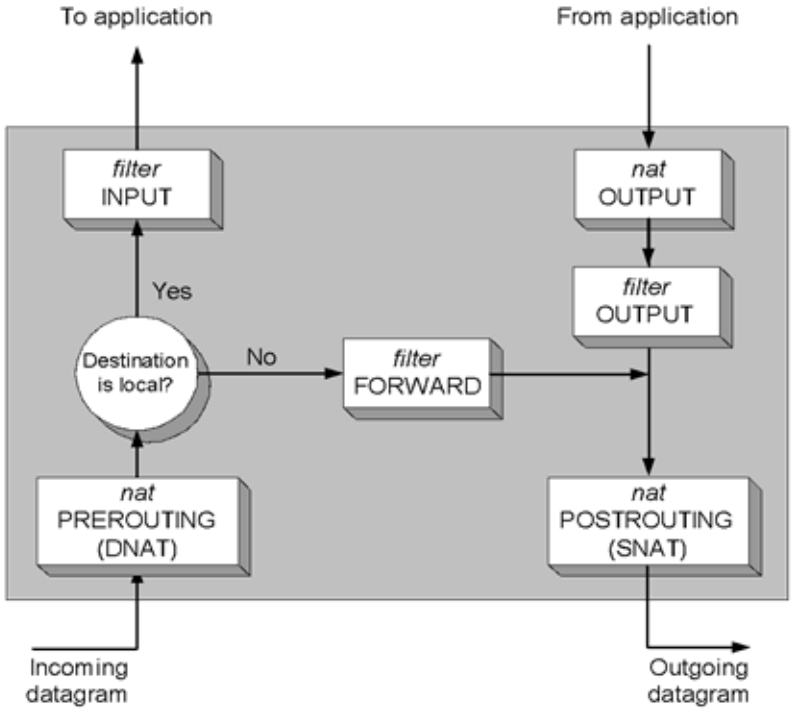
- **INPUT**
  - Perform a check on the packets destined to user-space
- **OUTPUT**
  - Perform a check on the packets originated from user-space
- **FORWARD**
  - Perform a check on the packets the traverse the firewall
  - By default, forwarding is disabled in Linux O.S.



NAT Table

Made of 3 chains:

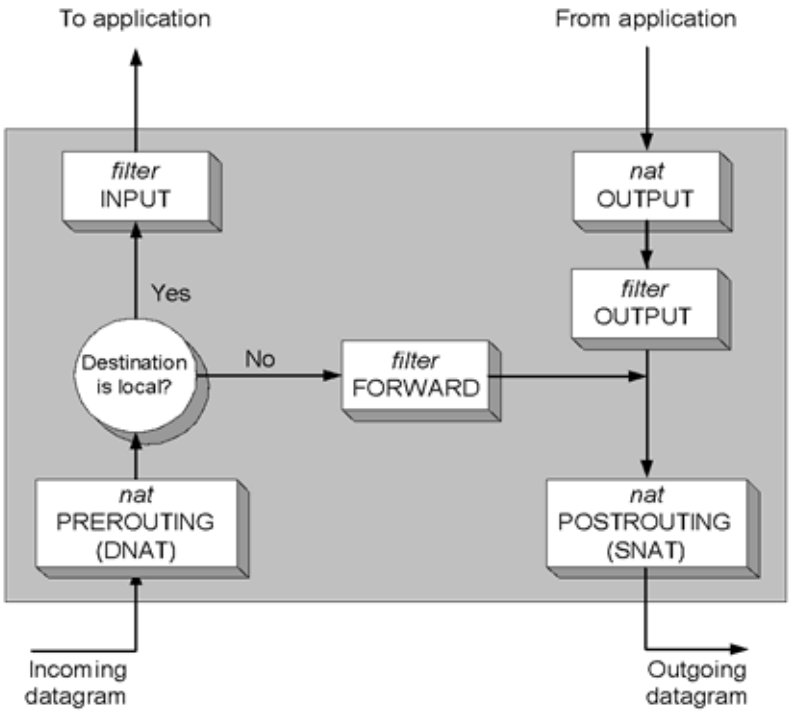
- **PREROUTING**
  - Perform some operations on the packet before the "routing decision" (**DNAT**)
- **OUTPUT**
  - Perform some operations on the packet generated by the firewall
- **POSTROUTING**
  - Perform some operations on the packet leaving the firewall (**SNAT**)



MANGLE Table

Made of 2 chains:

- **PREROUTING**
  - Perform some operations on the packet before the “routing decision”
- **OUTPUT**
  - Perform some operations on the packet generated by the firewall



IPTables: Rules

- Specified in each CHAIN
- Can be added
  - At the beginning of the rule-set
  - At the end of the rule-set
  - In a specific position in the rule-set
    - Rule order is important!!!
- A rule specifies matching criteria
  - Source and Destination IP addresses, ports
  - Source MAC Address
  - States
  - Invalid Packets
    - CRC error, fragments, ...
  - TCP flags
    - SYN, FIN, ACK, RST, URG, PSH, ALL, NONE
  - Rate limit
- What to do
  - Rule TARGET

IPTables – Rule Targets & Syntax

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- ACCEPT – let the packet through
- REJECT – sends ICMP error message
- DROP – reject, but don't send ICMP message
- MASQ – masquerade
- RETURN – end of chain; stop traversing this chain and resume the calling chain
- QUEUE – pass the packet to the user space
- (none) – rule's counters incremented and packet passed on (used for accounting)

iptables –t TABLE –A CHAIN –[i|o] IFACE –s w.x.y.z –d a.b.c.d –p PROT –m state --state STATE –j ACTION

- TABLE = nat | filter | mangle
- CHAIN = INPUT | OUTPUT | FORWARD | PREROUTING| POSTROUTING
- IFACE = eth0 | eth1 | ppp0 | …
- PROT = tcp | icmp | udp | …
- STATE = NEW | ESTABLISHED | RELATED | …
- ACTION = DROP | ACCEPT | REJECT | DNAT | SNAT |…

IPTable – Rule Syntax

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- Specifying IP addresses
  - Source: -s, --source or –src
  - Destination: -d, --destination or –dst
  - IP address can be specified in four ways.
    - (Fully qualified) host name (e.g., floyd, floyd.osis.cs.wright.edu
    - IP address (e.g., 127.0.0.1)
    - Group specification (e.g., 130.108.27.0/24)
    - Group specification (e.g., 130.108.27.0/255.255.255.0)
  - '–s ! IPaddress' and '–d ! IPaddress': Match address not equal to the given.
- Specifying an interface
  - Physical device for packets to come in
    - -i, --in-interface
    - -i eth0
  - Physical device for packets to go out
    - -o, --out-interface
    - -o eth3
  - INPUT chain has no output interface
  - OUPUT chain has no input interface
- Specifying a protocol
  - Protocol number/name
  - –p ! protocol

IPTables - Examples

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1. iptables --flush
   - » Delete all rules
2. iptables -A INPUT -i lo -j ACCEPT
   - » Accept all packets arriving on lo for local processes
3. iptables -A OUTPUT -o lo -j ACCEPT
4. iptables --policy INPUT DROP
   - » Unless other rules apply, drop all INPUT packets
5. iptables --policy OUTPUT DROP
6. iptables --policy FORWARD DROP
7. iptables -L -v -n
   - » List all rules, verbosely, using numeric IP addresses etc.
8. iptables -A INPUT -i eth1 -p tcp -s 192.168.17.1 --sport 1024:65535 -d 192.168.17.2 --dport 22 -j ACCEPT
   - Accept all TCP packets arriving on eth1 for local processes from 192.168.17.1 with any source port higher than 1023 to 192.168.17.2 and destination port 22
9. iptables -t nat -A PREROUTING -p TCP -i eth0 -d 128.168.60.12 --dport 80 -j DNAT --to-destination 192.168.10.2
   - Change the destination address of all TCP packets arriving on eth0 aimed at 128.168.60.12 port 80 to 192.168.10.2 port 80.

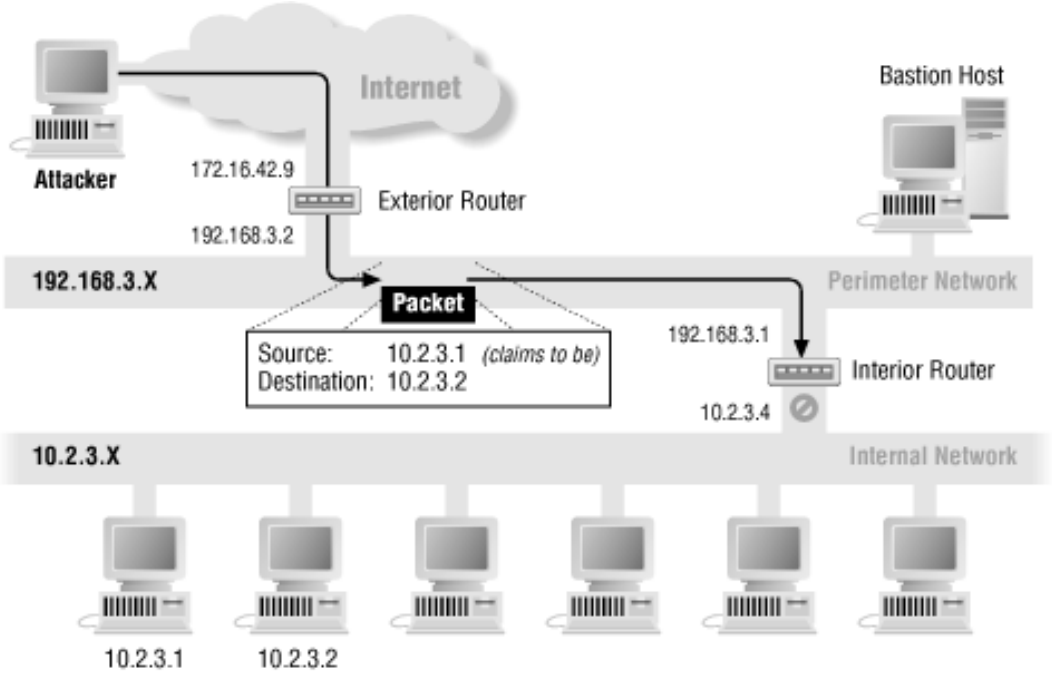
1. iptables –A INPUT –p tcp –s 0/0 –d 0/0 --dport 0:1023 –j REJECT
   - Reject all incoming TCP traffic destined for ports 0 to 1023

Iptables: one more example

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Enable (partial) “antispoof” protection

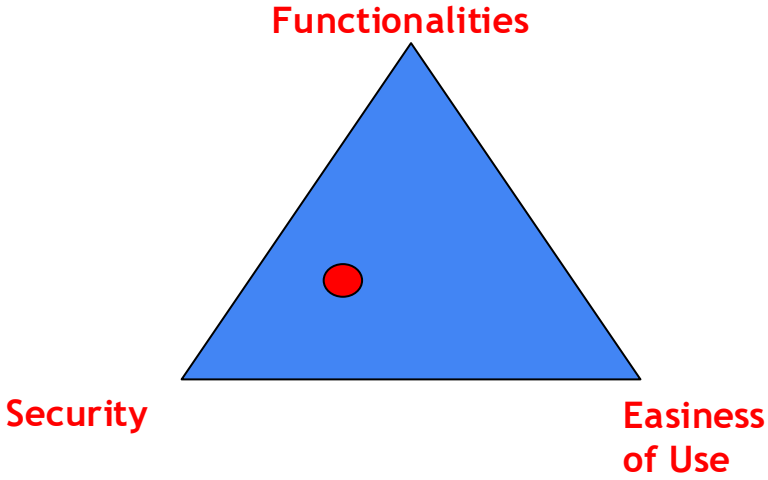
echo "1" > /proc/sys/net/ipv4/conf/all/rp_filter



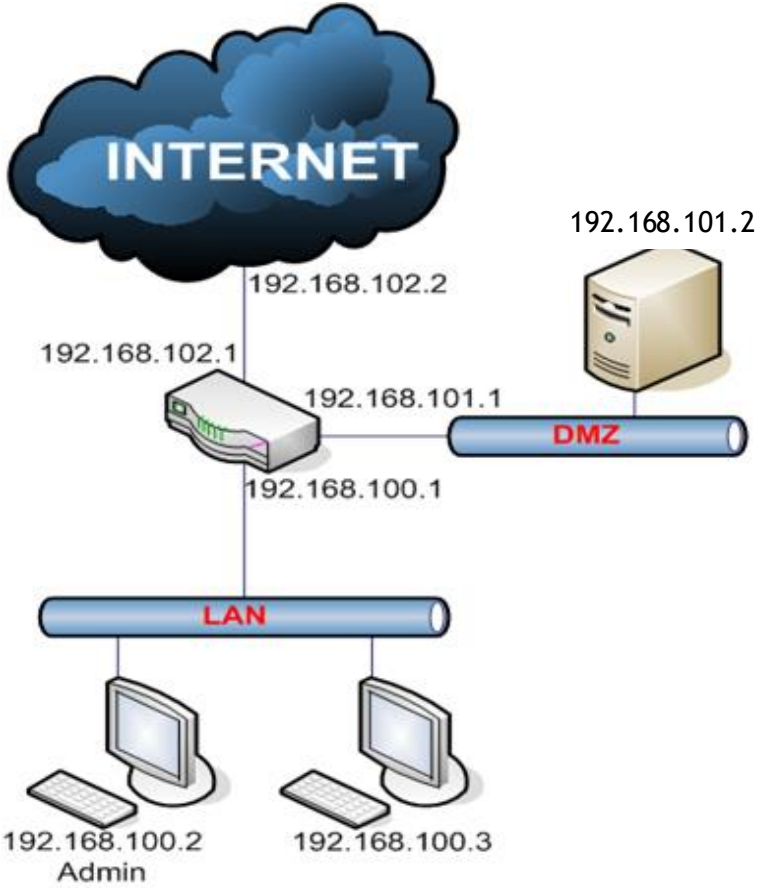
Lab

Best Practices

- Carefully define **Security Policies**
  - Least privilege
  - Fail safe
  - Weakest link
  - Defense in depth
- Do not be too restrictive
- Choose where to concentrate security
- Decide if internal users are trusted (no!)
- Think of possible "backdoors"
- Carefully design the security architecture
  - Host security
    - Anti-virus
    - IDS
    - Firewall
  - Network Security
    - Firewall
    - IDS
    - VPNs



Securing a single network



- **FIREWALL**
  - Linux (Iptables)
  - Commercial firewalls (e.g., checkpoint)
- INTRUSION DETECTION SYSTEM
  - Snort
  - Bro
  - Commercial IDSs
- Internal traffic protection
  - IPSec

Iptables

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- **Default Deny**
- **From LAN to FW**
  - SSH – only from Administrator
- **From LAN to DMZ**
  - SSH, Web, E-Mail, DNS, ICMP
- **From LAN to Ext**
  - SSH, Web, E-Mail, DNS, ICMP
- **From DMZ to LAN**
  - Nothing (really?)
- **From DMZ to FW**
  - Ping (?)
- **From DMZ to Ext**
  - SSH, Web, E-Mail, DNS
- **From Ext to FW**
  - Nothing
- **From Ext to LAN**
  - Nothing (really?)
- **From Ext to DMZ**
  - Web (Port Forwarding)

Iptables

```
#!/bin/sh
# Define interfaces
INTIF="eth0"
DMZIF="eth1"
EXTIF="eth2"
# Define interfaces addresses
INTIP="192.168.100.1"
DMZIP="192.168.101.1"
EXTIP="192.168.102.1"
# Define networks
INTNET="192.168.100.0/255.255.255.0"
DMZNET="192.168.101.0/255.255.255.0"
EXTNET="192.168.102.0/255.255.255.0"
# Note: in case of ADSL connection with dynamic IP address
# EXTIF="ppp0"
# EXTNET="0/0"
```

Iptables

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```
## Set Kernel parameters and load kernel module
# Enable IP forwarding (Essential!)
echo "1" > /proc/sys/net/ipv4/ip_forward
# Do not reply broadcast ping
echo "1" > /proc/sys/net/ipv4/icmp_echo_ignore_broadcasts
# Enale protection from bogus error messages
echo "1" > /proc/sys/net/ipv4/icmp_ignore_bogus_error_responses
# Do not accept icmp redirect
echo "0" > /proc/sys/net/ipv4/conf/all/accept_redirects
# Log “strange” packets
echo "1" > /proc/sys/net/ipv4/conf/all/log_martians
# Load kernel modules
modprobe ip_tables
modprobe iptable_nat
modprobe ip_conntrack
modprobe ipt_MASQUERADE
```

Iptables


```
# Reset rules and counters
iptables -t filter -F
iptables -t filter -X
iptables -t filter -Z
iptables -t mangle -F
iptables -t mangle -X
iptables -t mangle -Z
iptables -t nat -F
iptables -t nat -X
iptables -t nat -Z
```

Iptables


```
# Set default Policies
iptables -P INPUT DROP
iptables -P OUTPUT DROP
iptables -P FORWARD DROP
iptables -t nat -P POSTROUTING ACCEPT
iptables -t nat -P PREROUTING ACCEPT
iptables -t nat -P OUTPUT ACCEPT
iptables -t mangle -P OUTPUT ACCEPT
iptables -t mangle -P INPUT ACCEPT
iptables -t mangle -P FORWARD ACCEPT
iptables -t mangle -P POSTROUTING ACCEPT
iptables -t mangle -P PREROUTING ACCEPT

## Enable loopback traffic
```

Iptables

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```
## Traffic to/from firewall
# Address from which it is possible to access the firewall (ssh)
ADMIN="192.168.100.2"
## INPUT
# From internal network (ssh)
iptables -A INPUT -s $ADMIN -i $INTIF -p tcp --dport 22 -j ACCEPT
# Accept traffic related to already established connections
iptables -A INPUT -m state --state ESTABLISHED,RELATED -j ACCEPT
# Enable ICMP (e.g., ping) from INTNET and DMZ
iptables -A INPUT -p icmp -i $DMZIF -j ACCEPT
iptables -A INPUT -p icmp -i $INTIF -j ACCEPT
## OUTPUT
## (sanity check)
iptables -A OUTPUT -m state --state NEW,ESTABLISHED,RELATED -j ACCEPT
## LOCAL LAN NATTING (MASQUERADING)
iptables -t nat -A POSTROUTING -o $EXTIF -s $INTNET -j SNAT --to-source $EXTIP
```

Iptables

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```
## Traffic from LAN to DMZ and EXT,
iptables -A FORWARD -s $INTNET -i $INTIF -p tcp --dport 22 -j ACCEPT #(SSH)
iptables -A FORWARD -s $INTNET -i $INTIF -p tcp --dport 25 -j ACCEPT #(SMTP)
iptables -A FORWARD -s $INTNET -i $INTIF -p tcp --dport 110 -j ACCEPT #(POP3)
iptables -A FORWARD -s $INTNET -i $INTIF -p tcp --dport 143 -j ACCEPT #(IMAP2)
iptables -A FORWARD -s $INTNET -i $INTIF -p tcp --dport 995 -j ACCEPT #(POP3 over SSL)
iptables -A FORWARD -s $INTNET -i $INTIF -p tcp --dport 993 -j ACCEPT #(IMAP2 over SSL)
iptables -A FORWARD -s $INTNET -i $INTIF -p udp --dport 53 -j ACCEPT #(DNS)
iptables -A FORWARD -s $INTNET -i $INTIF -p tcp --dport 80  -j ACCEPT #(HTTP)
iptables -A FORWARD -s $INTNET -i $INTIF -p tcp --dport 443 -j ACCEPT #(HTTPS)
iptables -A FORWARD -s $INTNET -i $INTIF -p icmp --icmp-type echo-request -j ACCEPT #(outgoing Ping)
iptables -A FORWARD -s ! $INTNET -i ! $INTIF -p icmp --icmp-type echo-reply -j ACCEPT #(Ping Reply)
iptables -A FORWARD -s ! $INTNET -i ! $INTIF -p icmp --icmp-type time-exceeded -j ACCEPT #(Traceroute reply)
iptables -A FORWARD -s ! $INTNET -i ! $INTIF -p icmp --icmp-type destination-unreachable -j ACCEPT
```

Iptables

```
## Complete access from administrator to DMZ and EXT
 iptables -A FORWARD -s $ADMIN -j ACCEPT
## Traffic from DMZ to EXT,
## (e-mail, web, dns and ssh)
iptables -A FORWARD -s $DMZNET -i $DMZIF -d $EXTNET -o ! $INTIF -p tcp --dport 25 -j ACCEPT #(SMTP)
iptables -A FORWARD -s $DMZNET -i $DMZIF -d $EXTNET -o ! $INTIF -p udp --dport 53 -j ACCEPT #(DNS)
iptables -A FORWARD -s $DMZNET -i $DMZIF -d $EXTNET -o ! $INTIF -p tcp --dport 80  -j ACCEPT #(HTTP)
iptables -A FORWARD -s $DMZNET -i $DMZIF -d $EXTNET -o ! $INTIF -p tcp --dport 443 -j ACCEPT #(HTTPS)
## Traffic from EXT to DMZ (Web only)
## Nat of web server, with public access
# private webserver IP (dmz)
webserver="192.168.101.2"
# NAT 1-*
iptables -t nat -A PREROUTING -d $EXTIP -p tcp --dport 80 -j DNAT --to-destination $webserver:80
## Access to services in DMZ
iptables -A FORWARD -d $webserver -p tcp --dport 80 -j ACCEPT
```

Iptables

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```
# Accept traffic of established/related connections
iptables -A FORWARD -m state --state ESTABLISHED,RELATED -j ACCEPT
## SNAT of DMZ
iptables -t nat -A POSTROUTING -s $DMZNET -j SNAT --to-source $EXTIP


## LOG dropped packets
## unicast only (no broadcast/multicast)

iptables -A INPUT -m pkttype --pkt-type unicast -j LOG --log-prefix "INPUT DROP: "
iptables -A OUTPUT -m pkttype --pkt-type unicast -j LOG --log-prefix "OUTPUT DROP: "
iptables -A FORWARD -m pkttype --pkt-type unicast -j LOG --log-prefix "FORWARD DROP: "
```